Supplement to "Face selection for slowly switching Markov chains"

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Abstract

This supplement holds proofs for the paper "Face selection for slowly switching Markov chains". Section S1 proves the rows of the table of outcomes (Table 2 of the paper) that are proved by hand and not in the text, namely the path P_d and two triangles joined by a link. Section S2 gives the full proof of Proposition 4.14 of the paper, the directed 3-cycle, whose proof the paper only sketches. Sections S3 and S4 give the higher-order Laplace proof and the detailed next-term expansion, including the complete-graph coefficient calculation. Numbers with the prefix S refer to this supplement, and all other numbers refer to the paper.

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S1 Proofs for the table of outcomes

This section proves the rows P_d and Θ_ε of Table 2. In both rows Q is symmetric and the start has full support. So the admissible faces are the connected faces (Lemma 2.2), and the winners at τ are the connected faces that minimize $\Phi_\tau(F) = \tau\lambda_F + |F| - 1$ (Theorem 3.4). We use three facts about a connected face F. First, $-Q_F$ is symmetric, λ_F is its least eigenvalue, and an eigenvector of $-Q_F$ with positive entries belongs to λ_F (the second fact at the start of Appendix A of the paper). Second, $-Q_F$ is the Laplacian of the graph on F, which is positive semidefinite, plus the diagonal matrix of the rates out of F. So λ_F is the least value of

$$\sum_{\{i,j\}\subseteq F} q_{ij}(f_i - f_j)^2 + \sum_{i\in F} q_i^{\mathrm{out}}(F) f_i^2 \tag{S1.1}$$

over unit vectors $f \in \mathbb{R}^F$. Third, if $F' \subsetneq F$ is nonempty, then $\lambda_{F'} > \lambda_F$ (Proposition 3.1(d)).

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Each proof has two steps, namely the exit rates and then the lower-left boundary of the hull. For the second step we use the following lemma. It says that a chain of points with increasing switch values is its own lower-left boundary.

Lemma S1.1 (a chain of lines)**.** *Let $x_0 > x_1 > \cdots > x_m \ge 0$ and $y_0 < y_1 < \cdots < y_m$, and put $\ell_i(\tau) = \tau x_i + y_i$ and $t_i = (y_i - y_{i-1})/(x_{i-1} - x_i)$ for $1 \le i \le m$. Put $t_0 = 0$ and $t_{m+1} = \infty$, and suppose that $t_1 < t_2 < \cdots < t_m$. Then for $1 \le j \le m$ and every τ*

$$\ell_j(\tau) - \ell_{j-1}(\tau) = (x_{j-1} - x_j)(t_j - \tau). \tag{S1.2}$$

Let $0 \le i \le m$ and let τ be finite with $t_i \le \tau \le t_{i+1}$. Then $\ell_0(\tau) \ge \cdots \ge \ell_i(\tau) \le \cdots \le \ell_m(\tau)$. If also $t_i < \tau < t_{i+1}$, then all these inequalities are strict, so $\ell_i(\tau) < \ell_j(\tau)$ for every $j \ne i$.

Proof. The identity (S1.2) is the definition of t_j. If $j \le i$, then $t_j \le t_i \le \tau$, so $\ell_j(\tau) \le \ell_{j-1}(\tau)$. If $j > i$, then $\tau \le t_{i+1} \le t_j$, so $\ell_j(\tau) \ge \ell_{j-1}(\tau)$. If $t_i < \tau < t_{i+1}$, these inequalities are strict. □

S1.1 The path

Let P_d be the path with states $1, \dots, d$ and $q_{i,i+1} = q_{i+1,i} = 1$ for $1 \le i < d$. An *end-arc* of $k < d$ states is one of the two faces $\{1, \dots, k\}$ and $\{d-k+1, \dots, d\}$. For $k \ge 1$ put

$$\lambda'_k = 2 - 2\cos\frac{\pi}{2k+1}, \qquad \tau'_k = \frac{1}{\lambda'_{k-1} - \lambda'_k}\ (k \ge 2), \qquad D'_k = k + \lambda'_k \tau'_k\ (k \ge 2).$$

Lemma S1.2 (the numbers D'_k)**.** *(a) The sequence λ'_k is positive, strictly decreasing and strictly convex in $k \ge 1$. Hence $0 < \tau'_2 < \tau'_3 < \cdots$. Also $\tau'_2 = (1+\sqrt{5})/2$.*

(b) For every $k \ge 2$, $\frac{3}{2}k - \frac{1}{2} < D'_k < \frac{3}{2}k$.

Proof. (a) Let $g(x) = 2 - 2\cos(\pi/x)$, so that $\lambda'_k = g(2k+1)$. For $x \ge 2$,

$$g'(x) = -\frac{2\pi \sin(\pi/x)}{x^2} < 0, \qquad g''(x) = 2\pi\Big(\frac{\pi\cos(\pi/x)}{x^4} + \frac{2\sin(\pi/x)}{x^3}\Big) > 0.$$

So $\lambda'_{k-1} - \lambda'_k > \lambda'_k - \lambda'_{k+1} > 0$ for $k \ge 2$, and the τ'_k increase. Since $\lambda'_1 = 1$ and $\lambda'_2 = 2 - 2\cos\frac{\pi}{5} = (3-\sqrt{5})/2$, we get $\tau'_2 = 2/(\sqrt{5}-1) = (1+\sqrt{5})/2$.

(b) We turn the claim into two bounds on the ratio of two consecutive exit rates, and we prove each bound by comparing $\sin x$ with x. Put $a = \pi/(2(2k-1))$ and $b = \pi/(2(2k+1))$, so that $0 < b < a \le \pi/6$. Since $2 - 2\cos 2x = 4\sin^2 x$, we have $\lambda'_{k-1} = 4\sin^2 a$ and $\lambda'_k = 4\sin^2 b$. So the ratio $\rho = \lambda'_k/\lambda'_{k-1}$ equals $(\sin b/\sin a)^2$. By definition $D'_k - k = \lambda'_k/(\lambda'_{k-1} - \lambda'_k) = \rho/(1-\rho)$, which increases with ρ and takes the values $\frac{k-1}{2}$ and $\frac{k}{2}$ at $\rho = \frac{k-1}{k+1}$ and $\rho = \frac{k}{k+2}$. So the claim is

$$\frac{k-1}{k+1} < \rho < \frac{k}{k+2}.$$

If we replace each sine by its argument, then ρ becomes $(b/a)^2 = (2k-1)^2/(2k+1)^2$. This number lies between the two bounds and very close to the lower one. So the lower bound needs only $\sin b/\sin a > b/a$, and the upper bound needs the next term of the sine.

The lower bound. The function $\sin x/x$ decreases on $(0, \pi]$, so $\sin b/\sin a > b/a = (2k-1)/(2k+1)$. Then $\rho > (2k-1)^2/(2k+1)^2 > (k-1)/(k+1)$, where the last step holds because $(2k-1)^2(k+1) - (k-1)(2k+1)^2 = 2 > 0$.

The upper bound. Write $s(x) = \sin x/x$ and $y = (8k-2)/((k+2)(2k-1)^2)$. Then

$$\rho = \Big(\frac{2k-1}{2k+1}\Big)^2\Big(\frac{s(b)}{s(a)}\Big)^2 \qquad \text{and} \qquad \frac{k}{k+2} = \Big(\frac{2k-1}{2k+1}\Big)^2(1+y),$$

the second because $k(2k+1)^2-(k+2)(2k-1)^2 = 8k-2$. So we must show that $s(b)/s(a) < \sqrt{1+y}$. We reduce this in three steps to an inequality between two explicit functions of k.

Step 1. We remove the square root. We have $0 < y < 1$, because $(k+2)(2k-1)^2 \geq 12(2k-1) > 8k-2$, and $\sqrt{1+y} \geq 1+\frac{y}{3}$ for $0 \leq y \leq 3$. So it is enough to show that $s(b)/s(a) < 1+\frac{y}{3}$, that is

$$s(b) - s(a) < \tfrac{y}{3}\, s(a).$$

Step 2. We bound both sides with $1-\frac{x^2}{6} \leq s(x) \leq 1-\frac{x^2}{6}+\frac{x^4}{120}$, which holds for $x > 0$. On the right, $a \leq \pi/6$ gives $s(a) \geq 1-\frac{\pi^2}{216} > 0.954$, so $\frac{y}{3}s(a) > 0.318\, y$. On the left,

$$s(b)-s(a) \leq \frac{a^2-b^2}{6}+\frac{b^4}{120} = \frac{\pi^2 k}{3(4k^2-1)^2}+\frac{\pi^4}{1920(2k+1)^4}.$$

So it is enough to show that this is less than $0.318\, y$.

Step 3. We multiply both sides by $(2k-1)^2$, and on the left we use $k/(2k+1)^2 < 1/(4k)$ and $(2k-1)^2/(2k+1)^4 < 1/(4k^2)$. It is then enough to show that

$$\frac{\pi^2}{12k}+\frac{\pi^4}{7680k^2} < 0.318\,\frac{8k-2}{k+2}.$$

The left side decreases in k, and the right side, which is $0.318(8-\frac{18}{k+2})$, increases. At $k=2$ the left side is less than 0.415 and the right side is more than 1.11. So the inequality holds for every $k \geq 2$. □

Proposition S1.3 (the path)**.** *Let $d \geq 3$, let Q be the path P_d with unit rates, and let the start have full support. Put $K = \lfloor 2d/3 \rfloor$ and $\tau_{\text{full}} = \lceil d/3 \rceil / \lambda'_K$.*

(a) The connected faces are the sets of consecutive states. An end-arc of $k < d$ states has exit rate λ'_k. A set of k consecutive states that contains neither 1 nor d has exit rate $2-2\cos\frac{\pi}{k+1} > \lambda'_k$.

(b) We have $2 \leq K \leq d-1$ and $\tau'_2 < \cdots < \tau'_K < \tau_{\text{full}}$.

(c) The winners are the states 1 and d for $0 < \tau < \tau'_2$, the two end-arcs of k states for $\tau'_k < \tau < \tau'_{k+1}$ and $2 \leq k \leq K-1$, the two end-arcs of K states for $\tau'_K < \tau < \tau_{\text{full}}$, and P_d alone for $\tau > \tau_{\text{full}}$. No other face wins for any $\tau > 0$.

In words, the mode grows from an end of the path, one state at a time, until it covers $K = \lfloor 2d/3 \rfloor$ states (about two thirds of the path), and then it jumps to the whole path. The first step is at the golden ratio.

Idea of the proof. Growing an end-arc by one state lowers its exit rate by less and less, since λ'_k is convex. So the end-arcs alone win in the order of their size, and the arc of k states starts to win at τ'_k. The whole path enters when its exponent $d-1$ falls below the least exponent of the arcs. At τ'_k that least exponent is $D'_k - 1$, so the arcs of k states win before the whole path exactly when $D'_k < d$. Lemma S1.2(b) puts D'_k just below $\frac{3}{2}k$, and this gives $k \leq 2d/3$.

Proof. (a) A connected face of a path is a set of consecutive states. Let F have $k < d$ states. If $F = \{1,\dots,k\}$, then $-Q_F$ is tridiagonal with off-diagonal entries -1 and diagonal $(1,2,\dots,2)$. Put $\vartheta = \pi/(2k+1)$ and $r_j = \cos((2j-1)\vartheta/2)$ for $1 \leq j \leq k$. Then $r_j > 0$, since $(2j-1)\vartheta/2 < \pi/2$. With $r_0 = r_1$ and $r_{k+1} = \cos(\pi/2) = 0$ we have $r_{j-1}+r_{j+1} = 2\cos\vartheta\, r_j$ for $1 \leq j \leq k$, and this says that $-Q_F r = (2-2\cos\vartheta)r$. So $\lambda_F = \lambda'_k$, and the same holds for $\{d-k+1,\dots,d\}$ by the reflection $i \mapsto d+1-i$. If F contains neither 1 nor d, then $-Q_F = 2I - A$ with A the adjacency matrix of a path on k states.

Now $r_j = \sin(j\pi/(k+1))$ is a positive eigenvector with eigenvalue $2 - 2\cos\frac{\pi}{k+1}$, and this is larger than λ'_k because $\frac{\pi}{k+1} > \frac{\pi}{2k+1}$.

(b) Since $d \geq 3$ we have $2 \leq K \leq d-1$. The τ'_k increase by Lemma S1.2(a). Note that $d - K = \lceil d/3 \rceil$, so $\tau_{\text{full}} = (d-K)/\lambda'_K$. By Lemma S1.2(b), $D'_K < \frac{3}{2}K \leq d$, that is $\lambda'_K \tau'_K < d - K$. So $\tau'_K < \tau_{\text{full}}$.

(c) Write $\phi_k(\tau) = \tau\lambda'_k + k - 1$ for the exponent of an end-arc of k states, $1 \leq k \leq d-1$. First we apply Lemma S1.1 to the points $(\lambda'_k, k-1)$ for $1 \leq k \leq K$ followed by $(0, d-1)$, the point of P_d. Its switch values are $\tau'_2, \ldots, \tau'_K$ and $(d-K)/\lambda'_K = \tau_{\text{full}}$, which increase by (b). So on each of the intervals in (c) the face named there has a smaller exponent than the other end-arcs of at most K states and than P_d, and by (a) also than every set of at most K consecutive states that is not an end-arc.

It remains to show that a face F of k consecutive states with $K < k < d$ never wins. By (a) its exponent is at least $\phi_k(\tau)$. We show that $\phi_k(\tau) > \min(\phi_K(\tau), d-1)$ for every $\tau > 0$. Now we apply Lemma S1.1 to the points $(\lambda'_j, j-1)$ for $1 \leq j \leq d-1$, with switch values $\tau'_2 < \cdots < \tau'_{d-1}$. If $\tau < \tau'_{K+1}$, then $\tau < \tau'_j$ for $K < j \leq k$, and (S1.2) gives $\phi_k(\tau) > \phi_{k-1}(\tau) > \cdots > \phi_K(\tau)$. If $\tau \geq \tau'_{K+1}$, take j with $K+1 \leq j \leq d-1$ and $\tau'_j \leq \tau$, and with $\tau \leq \tau'_{j+1}$ if $j < d-1$. Then

$$\phi_k(\tau) \geq \phi_j(\tau) \geq \phi_j(\tau'_j) = D'_j - 1 > \tfrac{3}{2}j - \tfrac{3}{2} \geq \tfrac{3}{2}K \geq d - 1.$$

The first step is Lemma S1.1, the second holds since ϕ_j increases, the third is the definition of D'_j, the fourth is Lemma S1.2(b), the fifth is $j \geq K+1$, and the last holds because $3K \geq 2d-2$. So F never wins. The winners at τ are the admissible faces with the least exponent, and this gives (c). □

At a switch value the faces on both sides tie. On P_4 we have $K = 2$ and $D'_3 = 4.077 > 4$, so the end-arc of three states never wins. Its point lies only $\lambda'_3 - \frac{1}{2}\lambda'_2 = 0.0071$ to the right of the edge from the end-arcs of two states to P_4. The distance is small because the lower bound in Lemma S1.2(b) is nearly attained.

S1.2 Two triangles

Let $\varepsilon > 0$. The graph Θ_ε has states $1, \ldots, 6$, unit rates on the triangles $A = \{1,2,3\}$ and $B = \{4,5,6\}$, and one more link between 3 and 4 of rate ε. We call $1, 2, 5, 6$ the inner states and $\{1,2\}$, $\{5,6\}$ the inner edges. The graph has the symmetries $1 \leftrightarrow 2$, $5 \leftrightarrow 6$ and the mirror map $i \mapsto 7 - i$. Put

$$\begin{aligned} p_A(x) &= x^2 - (3+\varepsilon)x + \varepsilon, \\ p_4(x) &= x^3 - (5+2\varepsilon)x^2 + (6+6\varepsilon)x - 2\varepsilon, \\ p_5(x) &= x^3 - (4+2\varepsilon)x^2 + (3+4\varepsilon)x - \varepsilon, \end{aligned}$$

and $\varepsilon_0 = (64 - \sqrt{46})/60 = 0.95363$, $\varepsilon_1 = (64 + \sqrt{46})/60 = 1.17971$, the roots of $120\varepsilon^2 - 256\varepsilon + 135$.

Proposition S1.4 (two triangles joined by a link)**.** *Let the start have full support.*

(a) The two triangles have the same exit rate $\lambda_A = \lambda_B = \frac{1}{2}\big(3 + \varepsilon - \sqrt{\varepsilon^2 + 2\varepsilon + 9}\big)$, which is the smaller root of p_A, and $0 < \lambda_A < 1$.

(b) Among the connected faces of one, two and three states the least exit rates are 2, 1 and λ_A. They are attained exactly by the inner states, by the inner edges and by A and B. Put $\lambda_4 = \lambda_{\{1,2,3,4\}}$ and $\lambda_5 = \lambda_{\{2,\ldots,6\}}$. Every connected face of four states has exit rate at least λ_4, and every connected face of five states has exit rate λ_5. Here λ_4 and λ_5 are the least roots of p_4 and p_5, and $\lambda_5 < \lambda_4 < \lambda_A$.

(c) For every $\varepsilon > 0$, $\lambda_5 > \frac{1}{2}\lambda_4$, and $\lambda_4 > \frac{2}{3}\lambda_A$ if and only if $\varepsilon < \varepsilon_1$.

(d) Let $\varepsilon < \varepsilon_1$, and put

$$\tau_A = \frac{1}{1-\lambda_A} = \frac{1+\varepsilon+\sqrt{\varepsilon^2+2\varepsilon+9}}{4}, \qquad \tau^* = \frac{3}{\lambda_A} = \frac{3\big(3+\varepsilon+\sqrt{\varepsilon^2+2\varepsilon+9}\big)}{2\varepsilon}.$$

Then $1 < \tau_A < \tau^*$. The winners are the inner states for $0 < \tau < 1$, the inner edges for $1 < \tau < \tau_A$, the triangles A and B for $\tau_A < \tau < \tau^*$, and $[6]$ alone for $\tau > \tau^*$. No other face wins for any $\tau > 0$. As $\varepsilon \to 0$, $\tau_A = 1 + \frac{\varepsilon}{3} + O(\varepsilon^2)$ and $\tau^* = \frac{9}{\varepsilon} + 2 + O(\varepsilon)$.

(e) Let $\varepsilon > \varepsilon_1$. Then at $\tau = 3/\lambda_A$, where the triangles and $[6]$ have the same exponent, the face $\{1,2,3,4\}$ has a smaller exponent. So the list in (d) fails.

In words, for a weak link each triangle is a community that the chain rarely leaves. The mode sits on a triangle from $\tau \approx 1$ until $\tau \approx 9/\varepsilon$, and then it spreads over all six states at once. Once the link is stronger than ε_1 this list fails, because a triangle together with the far end of the link beats both the triangles and $[6]$ at the value of τ where they tie.

Idea of the proof. The exit rates of small faces are explicit, and those of the faces with four and five states are roots of cubics. The points of the inner states, the inner edges, the triangles and $[6]$ form a chain as in Lemma S1.1. The points of the faces with four and five states must lie strictly to the right of the edge from a triangle to $[6]$, and this is what (c) says. We prove (c) by continuity in ε. A change of sign needs a common root of two polynomials, hence a zero of a resultant, and the resultants are explicit polynomials in ε.

Proof. (a) On the vectors (a,a,b) the matrix $-Q_A$ acts as $(a,b) \mapsto (a-b,\ -2a+(2+\varepsilon)b)$, with characteristic polynomial p_A, and $(1,-1,0)$ is an eigenvector with eigenvalue 3. As $p_A(0) = \varepsilon > 0$ and $p_A(1) = -2 < 0$, the smaller root of p_A lies in $(0,1)$ and the other root is larger than 1. So the smaller root is λ_A. The mirror map gives $\lambda_B = \lambda_A$.

(b) By (S1.1), $\lambda_F \geq 1$ whenever every state of F has rate at least 1 out of F. We go through the connected faces by size. The states 3 and 4 have exit rate $2+\varepsilon$, and the inner states have 2. The connected pairs are the inner edges, with $\lambda = 1$, the pair $\{3,4\}$, with $\lambda = 2$, and $\{1,3\}$, $\{2,3\}$ and their mirror images, with $\lambda = \frac{1}{2}(4+\varepsilon-\sqrt{\varepsilon^2+4}) > 1$. The connected faces of three states are A, B and the sets $\{a,3,4\}$ and $\{3,4,b\}$ with $a \in \{1,2\}$ and $b \in \{5,6\}$, because a connected face that meets both triangles contains 3 and 4. In each of these four sets every state has rate at least 1 out of the set, so $\lambda \geq 1 > \lambda_A$. A connected face of four states meets both triangles, so it contains 3 and 4. So it is $\{1,2,3,4\}$, its mirror image $\{3,4,5,6\}$, or a set $\{a,3,4,b\}$, in which every state has rate 1 out of the set, so that $\lambda \geq 1$. A connected face of five states is the complement of an inner state (the complements of 3 and of 4 are disconnected), and these four faces are images of $\{2,\dots,6\}$ under the symmetries. By the third fact above, $\lambda_5 < \lambda_{\{3,4,5,6\}} = \lambda_4 < \lambda_A < 1$, since $\{2,\dots,6\} \supsetneq \{3,4,5,6\}$ and $\{1,2,3,4\} \supsetneq A$.

For the polynomials, note that $(1,-1,0,0)$ is an eigenvector of $-Q_F$ for $F = \{1,2,3,4\}$ with eigenvalue 3. On the vectors (a,a,b,c) the matrix acts as $(a,b,c) \mapsto (a-b,\ -2a+(2+\varepsilon)b-\varepsilon c,\ -\varepsilon b+(2+\varepsilon)c)$, with characteristic polynomial $(x-1)(x-2)(x-2-2\varepsilon)-2(x-2-\varepsilon) = p_4(x)$. So the eigenvalues of $-Q_F$ are 3 and the roots of p_4. As $\lambda_4 < 1$, it is the least root of p_4. Now let $F = \{2,\dots,6\}$. The vector that is 1 at 5, -1 at 6 and 0 elsewhere is an eigenvector of $-Q_F$ with eigenvalue 3. On the vectors with the value a at 2, b at 3, c at 4 and e at both 5 and 6, the matrix acts as

$$(a,b,c,e) \mapsto \big(2a-b,\ -a+(2+\varepsilon)b-\varepsilon c,\ -\varepsilon b+(2+\varepsilon)c-2e,\ -c+e\big).$$

Expanding the determinant along the first row gives the characteristic polynomial

$$(x-2)\big[(x-1)\big((x-2-\varepsilon)^2-\varepsilon^2\big)-2(x-2-\varepsilon)\big]-\big[(x-2-\varepsilon)(x-1)-2\big]=(x-3)\,p_5(x).$$

So the eigenvalues of $-Q_F$ are 3 (twice) and the roots of p_5. As $\lambda_5<1$, it is the least root of p_5. (Both characteristic polynomials are also checked by computer algebra, see Section S1.3.)

(c) Put $D_1=\lambda_5-\frac12\lambda_4$ and $D_2=\lambda_4-\frac23\lambda_A$. Both are continuous in $\varepsilon>0$, because the least eigenvalue of a symmetric matrix is continuous in its entries. If $D_1=0$, then $x=\lambda_5$ is a common root of $p_5(x)$ and $p_4(2x)$. If $D_2=0$, then $x=\lambda_4$ is a common root of $p_4(x)$ and $p_A(\frac32 x)$. A common root makes the resultant vanish, and an exact computation gives

$$\begin{aligned}\operatorname{Res}_x\big(p_5(x),p_4(2x)\big)&=-8\varepsilon^3(16\varepsilon^2+12\varepsilon+9),\\ \operatorname{Res}_x\big(p_A(\tfrac32 x),p_4(x)\big)&=-\tfrac{1}{16}\varepsilon^2(120\varepsilon^2-256\varepsilon+135).\end{aligned}$$

So D_1 has no zero in $\varepsilon>0$, and D_2 can vanish only at ε_0 and at ε_1. Hence D_1 has one sign on $(0,\infty)$, and D_2 has one sign on each of the intervals $(0,\varepsilon_0)$, $(\varepsilon_0,\varepsilon_1)$ and (ε_1,∞). A zero of the resultant does not force D_2 to vanish, because the common root need not be the least root of p_4. So we find the sign of D_2 for small ε, for large ε and at ε_0.

First let ε be small. By (a) and (b), $0<\lambda_5<\lambda_4<\lambda_A<\varepsilon$. Here $\lambda_5>0$ because $\lambda_5\ge0$ by (S1.1) and $p_5(0)=-\varepsilon\ne0$, and $\lambda_A<\varepsilon$ because $p_A(\varepsilon)=-2\varepsilon<0$. At $\varepsilon=0$ the cubics are $x(x-2)(x-3)$ and $x(x-1)(x-3)$, with the simple root 0. So the implicit function theorem gives one root $\frac{\varepsilon}{3}+O(\varepsilon^2)$ of each cubic near 0, and no other root near 0. So $\lambda_4=\frac{\varepsilon}{3}+O(\varepsilon^2)$ and $\lambda_5=\frac{\varepsilon}{3}+O(\varepsilon^2)$, and (a) gives $\lambda_A=\frac{\varepsilon}{3}+O(\varepsilon^2)$. Hence $D_1=\frac{\varepsilon}{6}+O(\varepsilon^2)$ and $D_2=\frac{\varepsilon}{9}+O(\varepsilon^2)$ are positive for small ε. So $D_1>0$ for every $\varepsilon>0$, and $D_2>0$ on $(0,\varepsilon_0)$.

Next, $D_2<0$ for large ε. Indeed $\lambda_A\to1$ as $\varepsilon\to\infty$ by (a). On the other hand take $f=(a,a,c,c)$ in (S1.1) for $\{1,2,3,4\}$. The value is $2(a-c)^2+2c^2$ and $|f|^2=2a^2+2c^2$, so $\lambda_4\le\min_{a,c}((a-c)^2+c^2)/(a^2+c^2)=(3-\sqrt5)/2<\frac12$ for every ε.

Now we show that $D_2>0$ at ε_0. Put $x_0=\frac{7}{40}$. Then

$$p_4(x_0)=\frac{57743}{64000}-\frac{809}{800}\varepsilon,\qquad p_A(\tfrac32 x_0)=\frac{59}{80}\varepsilon-\frac{4599}{6400}$$

are both negative for $0.8923<\varepsilon<0.9743$, and this interval contains ε_0. For $0\le x\le\frac12$ we have $p_4'(x)=3x^2-2(5+2\varepsilon)x+6+6\varepsilon\ge1+4\varepsilon>0$. So p_4 increases on $[0,x_0]$ and is negative there, and its least root λ_4 is larger than x_0. Also $p_A(\frac32 x_0)<0$ puts $\frac32 x_0$ between the roots of p_A, so $\lambda_A<\frac32 x_0$. Hence $\lambda_4>x_0>\frac23\lambda_A$ at ε_0.

So $D_2>0$ on $(0,\varepsilon_0]$. It has no zero in $(\varepsilon_0,\varepsilon_1)$, so $D_2>0$ on $(0,\varepsilon_1)$. It has no zero in (ε_1,∞) and is negative for large ε, so $D_2<0$ on (ε_1,∞). By continuity $D_2=0$ at ε_1. This proves (c).

(d) As $p_A(\frac34)=\frac{\varepsilon}{4}-\frac{27}{16}<0$ for $\varepsilon<\frac{27}{4}$, we have $\lambda_A<\frac34$. So $1-\lambda_A>0$ and $\lambda_A<3(1-\lambda_A)$, that is $1<\tau_A<\tau^*$. Apply Lemma S1.1 to the points $(2,0)$, $(1,1)$, $(\lambda_A,2)$ and $(0,5)$ of the inner states, the inner edges, the triangles and $[6]$. The switch values are 1, τ_A and τ^*. So on each interval in (d) the faces named there have a smaller exponent than the other three kinds, and by (b) than every other connected face of at most three states. Let F be a connected face of four or five states. By (b) and (c), for every $\tau>0$

$$\begin{aligned}\Phi_\tau(F)&\ge\tau\lambda_4+3>\tfrac23(\tau\lambda_A+2)+\tfrac13\cdot5 &&\text{if } |F|=4,\\ \Phi_\tau(F)&=\tau\lambda_5+4>\tfrac13(\tau\lambda_A+2)+\tfrac23\cdot5 &&\text{if } |F|=5,\end{aligned}$$

where the second line uses $\lambda_5>\frac12\lambda_4>\frac13\lambda_A$. Each right side is at least the smaller of the exponents $\tau\lambda_A+2$ of a triangle and 5 of $[6]$. So F never wins, and this gives the winners.

The closed forms follow from (a) by rationalizing, since $(\varepsilon^2+2\varepsilon+9)-(1+\varepsilon)^2=8$ and $(3+\varepsilon)^2-(\varepsilon^2+2\varepsilon+9)=4\varepsilon$. The expansions follow from $\sqrt{\varepsilon^2+2\varepsilon+9}=3+\frac{\varepsilon}{3}+O(\varepsilon^2)$.

(e) At $\tau=3/\lambda_A$ the triangles and $[6]$ have the exponent 5, and $\{1,2,3,4\}$ has $3\lambda_4/\lambda_A+3<5$ by (c). □

At $\varepsilon=1$, for example, $\tau_A=(1+\sqrt{3})/2$ and $\tau^*=6+3\sqrt{3}$.

S1.3 What was checked by computation

The proofs above use computer algebra in one place only, namely the two resultants in the proof of Proposition S1.4(c). The script `verify_examples.py` computes them in sympy. It also expands the characteristic polynomials of $-Q_F$ for $F=A$, $\{1,2,3,4\}$ and $\{2,\dots,6\}$ and finds $(x-3)p_A$, $(x-3)p_4$ and $(x-3)^2p_5$, and it recomputes $p_4(\frac{7}{40})$ and $p_A(\frac{21}{80})$ in rational arithmetic. The rest of its checks for these two rows are in floating point, and no proof rests on them. For $3\le d\le 9$ it computes the exit rates of all connected faces of P_d, compares those of the end-arcs with λ'_k, and finds the sizes of the winners as in Proposition S1.3(c). With 50 digits it confirms $\frac{3}{2}k-\frac{1}{2}<D'_k<\frac{3}{2}k$ for $2\le k\le 2000$. For $\varepsilon=0.3$, 1 and 1.1 it computes the exit rates of all 63 faces of Θ_ε and finds the least exit rates, the winners and the switch values of Proposition S1.4. For $\varepsilon=1.5$, 3, 4.4, 4.6 and 8 it finds that the faces $\{1,2,3,4\}$ and $\{3,4,5,6\}$ win on an interval of τ, which agrees with Proposition S1.4(e).

S2 The directed 3-cycle

This section proves Proposition 4.14 of the paper, whose statement we recall. Let $d=3$, $q_{12}=b>2$, $q_{23}=q_{31}=1$, all other rates 0, and $\mu_1=1$.

(i) The admissible faces are $\{1\}$, $\{1,2\}$ (reducible, $\lambda=1$) and $\{1,2,3\}$ (irreducible, $\lambda=0$). Exactly $M_{\{1\}}=(1-bh)^{N-1}$ and $M_{\{1,2\}}=bh(1-h)^{N-2}$, and the latter is attained only at $n=(1,N-1,0)$. For the full face, $\alpha^*=(1,b,b)/(1+2b)$, $\det\nabla^2 I=(1+2b)^4/(3b^2)$ and $K_{\{1,2,3\}}=(1+2b)^2/(2\sqrt{3}\,\pi b)$.

(ii) The equation $M_{\{1\}}=M_{\{1,2\}}$ has exactly one root $T_N^{(1)}$ with $bh<1$. It satisfies $(b-1)T+\log T=L-\log b+O(T^2/N)$, so $T_N^{(1)}=\frac{1}{b-1}[L-\log L+\log\frac{b-1}{b}]+O(\log L/L)$.

(iii) Every root $T\in[L/2,2L]$ of $M_{\{1,2\}}=M_{\{1,2,3\}}$ satisfies $T=L+\log\frac{2\sqrt{3}\,\pi b^2}{(1+2b)^2}+O(1/L)$, and these roots lie in an interval of length $O(1/L)$.

(iv) For large N and $T\le N^{1/3}$, the mode is $(N,0,0)$ for $T<T_N^{(1)}$, it is $(1,N-1,0)$ between $T_N^{(1)}$ and the smallest root in (iii), and it lies on the full face above the largest root in (iii).

Proof. (i) *The faces.* From state 1 the chain can only move along $1\to2\to3\to1$. So a path that starts at 1 has support $\{1\}$, $\{1,2\}$ or $\{1,2,3\}$, and these are the admissible faces. Inside $\{1,2\}$ the chain cannot return from 2 to 1, so this face is reducible with components $\{1\}$ and $\{2\}$, of exit rates b and 1, and $\lambda_{\{1,2\}}=1$. The only path with support $\{1\}$ stays at 1 for $N-1$ steps, which gives $M_{\{1\}}$. A path with support $\{1,2\}$ is $1^{n_1}2^{n_2}$, of probability $(1-bh)^{n_1-1}\,bh\,(1-h)^{n_2-1}$. Raising n_1 by one multiplies this by $(1-bh)/(1-h)<1$, so the maximum is at $n_1=1$ only, and it is $bh(1-h)^{N-2}$.

The full face. Here Q is a generator on $\{1,2,3\}$, so $\lambda=0$ and $r=\mathbf{1}$. The left vector l is the stationary law. In it the three flows $b\,l_1$, l_2 and l_3 around the cycle are equal, so $l=\alpha^*=(1,b,b)/(1+2b)$ and $(\mu\cdot r)(l\cdot\mathbf{1})=1$. By Proposition 4.2(a) it remains to find $\det\nabla^2 I$ at α^*, and then $K_{\{1,2,3\}}=\sqrt{\det\nabla^2 I}/(2\pi)$.

We first find the rate in closed form. By definition

$$I(\alpha) = \sum_i q_i \alpha_i - \inf_{u>0} \Big[b\alpha_1 \frac{u_2}{u_1} + \alpha_2 \frac{u_3}{u_2} + \alpha_3 \frac{u_1}{u_3} \Big].$$

The three ratios are positive numbers with product 1, and every such triple occurs. So by the AM–GM inequality the infimum is $3(b\alpha_1\alpha_2\alpha_3)^{1/3}$, and

$$I(\alpha) = \sum_i q_i \alpha_i - 3b^{1/3} g(\alpha), \qquad g(\alpha) = (\alpha_1\alpha_2\alpha_3)^{1/3}.$$

The first sum is linear, so the Hessian of I is $-3b^{1/3}$ times the Hessian of g. In all three coordinates, $\partial_i \partial_j g = -g\,\Omega_{ij}$ with $\Omega = \frac{1}{3}\operatorname{diag}(w)^2 - \frac{1}{9} w w^\top$ and $w = 1/\alpha$ (entrywise). In the coordinates (α_2, α_3) we have $\alpha_1 = 1-\alpha_2-\alpha_3$, so raising α_2 or α_3 moves α in the direction $v_2 = e_2 - e_1$ or $v_3 = e_3 - e_1$. Hence $\nabla^2 I$ is the 2×2 matrix with entries $3b^{1/3} g\, v_i^\top \Omega v_j$.

Now we evaluate at α^*. Put $s = 1 + 2b$ and $y = 1/b$. Then $3b^{1/3} g = 3b/s$ and $w = s(1, y, y)$, and a short computation gives

$$v_2^\top \Omega v_2 = v_3^\top \Omega v_3 = \tfrac{2}{9} s^2 (1 + y + y^2), \qquad v_2^\top \Omega v_3 = \tfrac{1}{9} s^2 (2 + 2y - y^2).$$

The determinant of this 2×2 matrix is a difference of two squares, namely

$$\frac{s^4}{81}\big[(2 + 2y + 2y^2)^2 - (2 + 2y - y^2)^2\big] = \frac{s^4}{81} \cdot 3y^2 (2 + y)^2 = \frac{s^6}{27 b^4},$$

since $2 + y = s/b$. Hence $\det \nabla^2 I = (3b/s)^2 s^6/(27b^4) = s^4/(3b^2)$ and $K_{\{1,2,3\}} = s^2/(2\sqrt{3}\,\pi b)$. The exact parts of (i) are checked in Lean.

(ii) We show that $D_1 = \log M_{\{1,2\}} - \log M_{\{1\}}$ increases in h. By (i), $D_1 = \log(bh) + (N-2)\log(1-h) - (N-1)\log(1-bh)$, so

$$D_1'(h) = \frac{1}{h} - \frac{N-2}{1-h} + \frac{b(N-1)}{1-bh} > 0 \qquad (bh < 1),$$

because $b > 1$ gives $b/(1-bh) > 1/(1-h)$. Also $D_1 \to -\infty$ as $h \to 0$ and $D_1 \to \infty$ as $bh \to 1$. So there is exactly one root (this is checked in Lean). To locate it we expand in $T = Nh$, which gives

$$D_1 = \log b + \log T - L + (b-1)T + O(T^2/N).$$

This is negative at $T = L/(2(b-1))$ and positive at $T = L/(b-1)$ for large N. So the root lies between them, and it satisfies the equation in (ii). Since the root is of order L, that equation gives $\log T = \log L - \log(b-1) + O(\log L/L)$, and putting this back gives the expansion.

(iii) By (i), $\log M_{\{1,2\}} = \log b + \log T - L - T + O(T^2/N)$. The full face satisfies (H), and Theorem 4.6 with $k = 3$ and $\lambda = 0$ gives $\log M_{\{1,2,3\}} = -2L + \log T + \log K_{\{1,2,3\}} + O(1/T + T^2/N)$. So on $[L/2, 2L]$

$$\log M_{\{1,2\}} - \log M_{\{1,2,3\}} = L - T + \log b - \log K_{\{1,2,3\}} + O(1/L).$$

At a root the left side is 0, which gives the formula in (iii) with the value of $K_{\{1,2,3\}}$ from (i). Two roots then differ by $O(1/L)$.

(iv) We compare the three face maxima on three ranges of T. Note first that Lemma 3.2 with $\theta = q$ gives $M_{\{1,2,3\}} \le C(T/N)^2$ for $T \le N^{1/3}$.

Let $T < T_N^{(1)}$. Then $M_{\{1\}} > M_{\{1,2\}}$ by (ii). Also $T < L/(b-1)$ by (ii), so $M_{\{1\}} = (1-bh)^{N-1} \ge N^{-b/(b-1)-o(1)}$. This is larger than $C(T/N)^2$ because $b > 2$ gives $b/(b-1) < 2$. So the mode is $(N, 0, 0)$.

Let $T > T_N^{(1)}$. Then $M_{\{1,2\}} > M_{\{1\}}$ by (ii), and we compare $\{1,2\}$ with the full face. For $T \le L/2$ the bound on $M_{\{1,2,3\}}$ gives $\log M_{\{1,2\}} - \log M_{\{1,2,3\}} \ge L - T - \log T - C > 0$. On $[L/2, 2L]$ the difference in (iii) is continuous in T, positive at $L/2$ and negative at $2L$. So it is positive below its smallest root and negative above its largest root. For $2L \le T \le N^{1/3}$ the same expansions give a difference of at most $-L + C < 0$. Since the maximum of $M_{\{1,2\}}$ is attained only at $(1, N-1, 0)$, this proves (iv). □

S3 The higher-order Laplace estimate

This section proves Lemma B.2 of the paper. Its hypotheses, coefficient formula and notation are stated in Appendix B.1 there.

In words, c_1 is the mean of an explicit polynomial of degree 6 under the Gaussian law with covariance $\mathcal{J}^{-1}$. As a formal check of the coefficients in (B.1), take the integrand $e^{T(e^{\mathrm{i}\varphi} - 1 - \mathrm{i}\varphi)}$ with $m = 1$, $a = 1$ and integer T. Its integral over $[-\pi, \pi]$ is $2\pi e^{-T} T^T / T!$, and Stirling's formula gives $(2\pi/T)^{1/2}(1 - \frac{1}{12T} + O(T^{-2}))$. Here $\mathcal{J} = 1$, $\Psi_{111} = -\mathrm{i}$ and $\Psi_{1111} = 1$, so (B.1) gives $c_1 = \frac{1}{8} - \frac{1}{12} - \frac{1}{8} = -\frac{1}{12}$ as well. The check is only formal, because $\Psi = e^{\mathrm{i}\varphi} - 1 - \mathrm{i}\varphi$ is not 2π-periodic. So it is not a function on the torus, and the lemma does not cover it as stated. Only $e^{T\Psi}$ is periodic, for integer T.

Idea of the proof. After the substitution $\varphi = x/\sqrt{T}$ the integrand is a Gaussian weight times a power series in $T^{-1/2}$ whose coefficients are polynomials in x. The coefficients of the odd powers of $T^{-1/2}$ are odd polynomials and integrate to 0. The coefficient of T^{-1} has the Gaussian mean c_1, by Wick's formula.

Proof. We follow the proof of Lemma B.1 with its notation, and expand further. Now B_Ψ and B_a are the maxima over $|\beta| \le 6$ and $|\beta| \le 4$. The localization and the scaling are as before. The integral over $\|\varphi\| \ge \varepsilon$ is at most $(2\pi)^m B_a e^{T(\Psi(\alpha,0) - \eta)} \le C T^{-2-m/2} e^{T\Psi(\alpha,0)}$.

Taylor expansion. Put $s = T^{-1/2}$, so $0 < s \le 1$. For $j = 3, 4, 5$ let $\mathcal{P}_j(x)$ be the part of degree j of the Taylor polynomial of $\Psi(\alpha, \cdot)$ at 0. So $\mathcal{P}_3 = P$ and $\mathcal{P}_4(x) = \frac{1}{24}\Psi_{ijkl} x^i x^j x^k x^l$. For $j = 1, 2, 3$ let $A_j(x)$ be the part of degree j of the Taylor polynomial of $a(\alpha, \cdot)$, so $A_1(x) = a_i x^i$ and $A_2(x) = \frac{1}{2} a_{ij} x^i x^j$. Taylor's theorem to order 6 for Ψ and to order 4 for a gives, on the cube $\|x\| \le \varepsilon\sqrt{T}$,

$$w = w_0 + \omega_1, \qquad w_0 = s\mathcal{P}_3 + s^2\mathcal{P}_4 + s^3\mathcal{P}_5, \qquad a(\alpha, sx) = a_0 + sA_1 + s^2 A_2 + s^3 A_3 + \omega_2,$$

with $|\omega_1| \le C s^4 |x|^6$ and $|\omega_2| \le C s^4 |x|^4$. Let $\mathsf{p}(v) = 1 + v + \frac{1}{2}v^2 + \frac{1}{6}v^3$. We compare e^w with $\mathsf{p}(w)$, then with $\mathsf{p}(w_0)$, and then with

$$Y = 1 + s\mathcal{P}_3 + s^2\big(\mathcal{P}_4 + \tfrac{1}{2}\mathcal{P}_3^2\big) + s^3\big(\mathcal{P}_5 + \mathcal{P}_3\mathcal{P}_4 + \tfrac{1}{6}\mathcal{P}_3^3\big),$$

which collects the terms of $\mathsf{p}(w_0)$ of degree at most 3 in s. First, $|e^v - \mathsf{p}(v)| \le \frac{1}{24}|v|^4 e^{|v|}$, and we know that $|w| \le C s |x|^3$ and $|w| \le \frac{1}{4}\sigma_{\min}|x|^2$. So $|e^w - \mathsf{p}(w)| \le C s^4 |x|^{12} e^{\sigma_{\min}|x|^2/4}$. Second, $\mathsf{p}(w) - \mathsf{p}(w_0) = \omega_1\big[1 + \frac{1}{2}(w + w_0) + \frac{1}{6}(w^2 + w w_0 + w_0^2)\big]$, and $|w|$ and $|w_0|$ are at most $C(1 + |x|)^5$, by $|w| \le C s |x|^3$ and $s \le 1$. So $|\mathsf{p}(w) - \mathsf{p}(w_0)| \le C s^4 (1 + |x|)^{16}$. Third, $\mathsf{p}(w_0) - Y$ is a sum of terms s^j with $4 \le j \le 9$ times polynomials in x of degree at most 15 with bounded coefficients, so $|\mathsf{p}(w_0) - Y| \le C s^4 (1 + |x|)^{15}$. Together,

$$|e^w - Y| \le C s^4 (1 + |x|)^{16} e^{\sigma_{\min}|x|^2/4}.$$

Now write $e^w a = Y(a_0 + sA_1 + s^2 A_2 + s^3 A_3) + Y\omega_2 + (e^w - Y)a$. Here $|Y| \le C(1 + |x|)^9$ and $|a| \le B_a$. In the first product the terms of degree at least 4 in s are at most $C s^4 (1 + |x|)^{12}$. Hence

$$e^w a(\alpha, sx) = a_0 + s\,\mathcal{Q}_1(x) + s^2\mathcal{Q}_2(x) + s^3\mathcal{Q}_3(x) + \omega(x), \qquad |\omega(x)| \le C T^{-2} (1 + |x|)^{16} e^{\sigma_{\min}|x|^2/4},$$

where

$$\mathcal{Q}_1 = a_0\mathcal{P}_3 + A_1, \qquad \mathcal{Q}_2 = a_0\big(\mathcal{P}_4 + \tfrac{1}{2}\mathcal{P}_3^2\big) + A_1\mathcal{P}_3 + A_2,$$
$$\mathcal{Q}_3 = a_0\big(\mathcal{P}_5 + \mathcal{P}_3\mathcal{P}_4 + \tfrac{1}{6}\mathcal{P}_3^3\big) + A_1\big(\mathcal{P}_4 + \tfrac{1}{2}\mathcal{P}_3^2\big) + A_2\mathcal{P}_3 + A_3.$$

Every term of $\mathcal{Q}_1$ and of $\mathcal{Q}_3$ has odd degree, so $\mathcal{Q}_1$ and $\mathcal{Q}_3$ are odd functions.

Integration. As before, ω contributes at most CT^{-2} to the integral over the cube, since $(1+|x|)^{16}e^{-\sigma_{\min}|x|^2/4}$ is integrable. Extending the integrals of the four polynomial terms to $\mathbb{R}^m$ changes them by at most $Ce^{-\sigma_{\min}\varepsilon^2 T/4} \le CT^{-2}$. On $\mathbb{R}^m$ the odd terms $\mathcal{Q}_1$ and $\mathcal{Q}_3$ integrate to 0 against $e^{-\frac{1}{2}x^\top \mathcal{J}x}$. This is why the error is $O(T^{-2})$ and not $O(T^{-3/2})$. So the integral over the cube is

$$e^{T\Psi(\alpha,0)}T^{-m/2}\Big[(2\pi)^{m/2}\det\mathcal{J}^{-1/2}\Big(a_0 + \frac{\mathbb{E}\mathcal{Q}_2(x)}{T}\Big) + O(T^{-2})\Big],$$

where x is a centered Gaussian vector with $\mathbb{E}[x^i x^j] = \mathcal{J}^{ij}$. As in the proof of Lemma B.1, the errors are $O(T^{-2})$ relative to the main factor, because $\det\mathcal{J} \le (mB_\Psi)^m$.

It remains to show that $\mathbb{E}\mathcal{Q}_2 = c_1$. By Wick’s formula a Gaussian moment of order 4 or 6 is the sum, over the three or fifteen ways to pair the indices, of the products of the covariances of the pairs. The derivatives are symmetric in their indices. So $\mathbb{E}[A_2] = \frac{1}{2}a_{ij}\mathcal{J}^{ij}$, $\mathbb{E}[A_1\mathcal{P}_3] = \frac{1}{6}\cdot 3\,a_i\Psi_{jkl}\mathcal{J}^{ij}\mathcal{J}^{kl}$ and $\mathbb{E}[\mathcal{P}_4] = \frac{1}{24}\cdot 3\,\Psi_{ijkl}\mathcal{J}^{ij}\mathcal{J}^{kl}$. In $\mathbb{E}[\mathcal{P}_3^2] = \frac{1}{36}\Psi_{ijk}\Psi_{lrs}\mathbb{E}[x^ix^jx^kx^lx^rx^s]$, six pairings join each index of the first factor to an index of the second, and each gives $\Psi_{ijk}\Psi_{lrs}\mathcal{J}^{il}\mathcal{J}^{jr}\mathcal{J}^{ks}$. The other nine pair two indices inside each factor and join the remaining two, and each gives $\Psi_{ijk}\Psi_{lrs}\mathcal{J}^{ij}\mathcal{J}^{kl}\mathcal{J}^{rs}$. So $\frac{1}{2}\mathbb{E}[\mathcal{P}_3^2]$ gives the last two terms of (B.1). Finally c_1 is a polynomial in a_0, in the derivatives at $\varphi = 0$ and in the entries of $\mathcal{J}^{-1}$, which are continuous on $\mathcal{K}$. □

S4 Proof of the next-term expansion

This section proves Proposition 4.15 of the paper. The coefficient $c(\alpha)$ is defined in (B.2), using (B.1). All hypotheses and notation are those of the proposition and Appendix B.2.

Throughout, F satisfies (H). If $F = \{i\}$, then (2.2) gives $\log M_F = \log\mu_i - q_iT + O(T^2/N)$, which is (4.7) with $c_F = 0$. So let $k \ge 2$ until the proof of (iii).

Part (i). We use the proof of Theorem 4.4 in Appendix A. It shows that Ψ and χ in (A.6) satisfy the hypotheses of Lemma B.1 with $m = k-1$, and that their φ-derivatives of every order are jointly continuous on $\mathcal{K}\times\mathbb{T}^{k-1}$. So Lemma B.2 applies with $a = \chi$. Let $c_1(\alpha)$ be the function (B.1) for this Ψ and $a = \chi$, with $\mathcal{J}(\alpha) = -\nabla_\varphi^2\Psi(\alpha,0)$, and use the definition (B.2) of the paper. Then the bracket $\chi(\alpha,0) + O(1/T)$ in the last step of that proof becomes $\chi(\alpha,0) + c_1(\alpha)/T + O(T^{-2})$. Since $\chi(\alpha,0)$ is bounded below on $\mathcal{K}$, this is $\chi(\alpha,0)(1 + c(\alpha)/T + O(T^{-2}))$, and the rest of that proof gives the expansion in (i). By the regularity step of that proof, Ψ and χ are real-analytic in (α,φ) near $\{\alpha_0\}\times\mathbb{T}^{k-1}$ for every $\alpha_0 \in \Delta_F^\circ$. So their φ-derivatives at $\varphi = 0$ are real-analytic in α, and so are the entries of $\mathcal{J}(\alpha)^{-1}$ and c. Finally (i) gives $c(\alpha) = \lim_{T\to\infty} T\big[f_T(\alpha)T^{-(k-1)/2}C_F(\alpha)^{-1}e^{TI_F(\alpha)} - 1\big]$. So c is real, and it does not depend on which state of F we number 1.

Part (ii)*, the plan.* We write $\log P_N(n)$ as a fixed term plus a function G_T of $\alpha = n/N$, and we maximize G_T. Far from α_F^*, at distance more than a large multiple of $1/T$, the quadratic growth of I_F makes G_T smaller than $(c_F - 1)/T$. At distance at most that multiple of $1/T$, the function G_T is a constant plus a linear term minus a positive definite quadratic form, up to an error $O(T^{-2})$. Completing the square shows that this quadratic model has the maximum c_F/T, and that it takes it exactly at the point $\alpha' = \alpha_F^{*\prime} + \Sigma_F\gamma_F/T$. So $G_T \le c_F/T + O(T^{-2})$, which gives the upper bound. A lattice point next to that point gives the lower bound.

Part (ii)*, the constants.* Write $\Sigma = \Sigma_F$, $\gamma = \gamma_F$ and $c^* = c(\alpha_F^*)$, and put

$$X = N^{-(k-1)} T^{(k-1)/2} K_F e^{-T\lambda_F}, \qquad x = \alpha' - \alpha_F^{*\prime}.$$

By Propositions 3.1(a) and 4.2(a) and by (i), the functions I_F, $\log C_F$ and c are real-analytic near α_F^*, and I_F has its minimum λ_F at α_F^* with Hessian Σ^{-1}. So there are a closed ball $U = \{|x| \le \rho_0\} \subset \Delta_F^\circ$ and constants $c_U > 0$ and L_1 such that on U

$$\Big|\log \frac{C_F(\alpha)}{K_F} - \gamma \cdot x\Big| \le L_1|x|^2, \qquad \Big|I_F(\alpha) - \lambda_F - \tfrac{1}{2} x^\top \Sigma^{-1} x\Big| \le L_1|x|^3,$$
$$|c(\alpha) - c^*| \le L_1|x|, \qquad |c(\alpha)| \le L_1, \qquad I_F(\alpha) - \lambda_F \ge c_U|x|^2.$$

Put $L_2 = |\gamma| + L_1\rho_0$, so that $|\log(C_F(\alpha)/K_F)| \le L_2|x|$ on U. Fix ρ_1 so large that

$$\rho_1 \ge \frac{L_2}{c_U}, \qquad \rho_1 \ge |\Sigma\gamma| + k, \qquad c_U\rho_1^2 - L_2\rho_1 - L_1 \ge |c_F| + 1.$$

For $\alpha \in U$ let

$$G_T(\alpha) = \log \frac{C_F(\alpha)}{K_F} - T\big(I_F(\alpha) - \lambda_F\big) + \frac{c(\alpha)}{T}.$$

By (i) with $\mathcal{K} = U$ and Lemma 4.5, there are T_0 and N_0 such that for $N \ge N_0$, $T_0 \le T \le N^{1/3}$ and $\operatorname{supp} n = F$ with $\alpha = n/N \in U$,

$$\log \frac{P_N(n)}{X} = G_T(\alpha) + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big). \tag{S4.1}$$

Here we used $\log(1 + c/T + O(T^{-2})) = c/T + O(T^{-2})$, which holds because $|c| \le L_1$ on U. We take $T_0 \ge \rho_1/\rho_0$, and we increase T_0 and N_0 finitely many times below.

Part (ii)*, two bounds for G_T.* First let $\alpha \in U$ with $|x| \ge \rho_1/T$. Then $G_T(\alpha) \le L_2|x| - c_U T|x|^2 + L_1/T$. The function $v \mapsto L_2 v - c_U T v^2$ decreases for $v \ge L_2/(2c_U T)$, and ρ_1/T lies in this range. So

$$G_T(\alpha) \le \frac{L_2\rho_1 - c_U\rho_1^2 + L_1}{T} \le \frac{c_F - 1}{T} \qquad (\alpha \in U,\ |x| \ge \rho_1/T). \tag{S4.2}$$

Next let $|x| \le \rho_1/T$, and put $y = x - \Sigma\gamma/T$. Then $\alpha \in U$, and the bounds on U give $G_T(\alpha) = \gamma \cdot x - \frac{T}{2} x^\top \Sigma^{-1} x + c^*/T + \epsilon$ with $|\epsilon| \le L_1|x|^2 + L_1 T|x|^3 + L_1|x|/T \le L_1(\rho_1^2 + \rho_1^3 + \rho_1)T^{-2}$. Completing the square gives $\gamma \cdot x - \frac{T}{2} x^\top \Sigma^{-1} x = \frac{1}{2T}\gamma^\top\Sigma\gamma - \frac{T}{2} y^\top \Sigma^{-1} y$. So

$$G_T(\alpha) = \frac{c_F}{T} - \frac{T}{2} y^\top \Sigma^{-1} y + O\Big(\frac{1}{T^2}\Big) \qquad (|x| \le \rho_1/T), \tag{S4.3}$$

uniformly in x. Since Σ^{-1} is positive definite, (S4.2) and (S4.3) give $G_T \le c_F/T + O(T^{-2})$ on U.

Part (ii)*, the upper bound.* The part of the proof of Theorem 4.6 that treats compositions outside U uses only that U is a closed ball around α_F^* inside Δ_F°. So there is $\eta > 0$ with $P_N(n) \le X \cdot O(T^{(k-1)/2} e^{-2\eta T})$ for $\operatorname{supp} n = F$ and $n/N \notin U$. After increasing T_0 this is at most $X e^{-\eta T} \le X e^{(c_F - 1)/T}$. For $n/N \in U$ we use (S4.1) and the bound on G_T. Hence $\log(M_F/X) \le c_F/T + O(T^{-2} + T^2/N)$.

Part (ii)*, the lower bound.* Let α^0 be the point with $x = \Sigma\gamma/T$. Let n_i^0 be an integer nearest to $N\alpha_i^0$ for $i \ge 2$, and $n_1^0 = N - \sum_{i \ge 2} n_i^0$. Then $|n_i^0 - N\alpha_i^0| \le k$ for every i. Since $\alpha^0 \to \alpha_F^*$ as $T \to \infty$ and every $\alpha_{F,i}^*$ is positive, n^0 has support F after increasing T_0 and N_0. For $\alpha = n^0/N$ we have $|y| \le k/N$ and $|x| \le |\Sigma\gamma|/T + k/N \le \rho_1/T$, because $T \le N$. So (S4.3) gives $G_T(n^0/N) \ge c_F/T - O(T/N^2 + T^{-2})$, and (S4.1) gives $\log(M_F/X) \ge \log(P_N(n^0)/X) \ge c_F/T - O(T^{-2} + T^2/N)$. This bound is sufficient for the claimed remainder.

The two bounds give (4.7) for $N \ge N_0$ and $T_0 \le T \le N^{1/3}$. For $1 \le T \le T_0$ it follows from (4.3), since c_F/T and $1/T$ are $O(T^{-2})$ there, with a constant that depends on T_0. This proves (ii).

Part (iii). Now F and G are as in Theorem 4.7, and $k_G = 1$ is allowed. Let $T_N \in [1, N^{1/3}]$ be a zero of D_N. The proof of Theorem 4.7(ii) shows that $\frac{\Delta}{2\delta} L < T_N \le \frac{2\Delta}{\delta} L$ for large N. Subtracting (4.7) for G from (4.7) for F at $T = T_N$ gives

$$0 = -\Delta L + \frac{\Delta}{2} \log T_N + \delta T_N + \log \frac{K_F}{K_G} + \frac{c_F - c_G}{T_N} + O\Big(\frac{1}{T_N^2} + \frac{T_N^2}{N}\Big).$$

We multiply by T_N. The error becomes $O(1/T_N + T_N^3/N) = O(1/L)$, which is (4.8).

Part (iv), *a symmetric pair*. Let $F = \{i, j\}$ with $q_{ij} = q_{ji} = a > 0$, $q_i = q_j = q$ and $\mu_i = \mu_j > 0$. Then F satisfies (H). We first prove (4.9), without the Laplace lemma. Let $1 \le T \le N^{1/3}$ and let N be so large that $qh \le \frac{1}{2}$ and $I + hQ$ is stochastic. By Proposition 5.1, $M_F = \mu_i (1-qh)^{N-1} S_N(u)$ with $u = ah/(1-qh)$. Put $z = Nu = aT/(1-qh)$. Then $aT \le z \le 2aT$ and $0 \le z - aT \le 2aqT^2/N$. By (2.1),

$$\log S_N(u) = \log \frac{2}{N} + \log \psi(z) + O\Big(\frac{T^2}{N}\Big),$$

since $z(z+2)/(2N) = O(T^2/N)$ and this is at most $\frac{1}{2}$ for large N. By [1, Eq. 10.40.1], as $z \to \infty$, $I_0(z) = e^z (2\pi z)^{-1/2} (1 + \frac{1}{8z} + O(z^{-2}))$ and $I_1(z) = e^z (2\pi z)^{-1/2} (1 - \frac{3}{8z} + O(z^{-2}))$. So $I_0(z) + I_1(z) = 2e^z (2\pi z)^{-1/2} (1 - \frac{1}{8z} + O(z^{-2}))$ and

$$\log \psi(z) = z + \tfrac{1}{2} \log z + \tfrac{1}{2} \log \frac{2}{\pi} - \frac{1}{8z} + \rho(z), \qquad |\rho(z)| \le C_a z^{-2} \quad \text{for } z \ge a.$$

The bound on ρ holds for all $z \ge a$, and not only for large z, because $\rho(z) = O(z^{-2})$ as $z \to \infty$ and ρ is continuous on $[a, \infty)$. The function $z + \frac{1}{2} \log z - \frac{1}{8z}$ has a bounded derivative on $[a, \infty)$. So replacing z by aT in it changes it by $O(T^2/N)$, and

$$\log \psi(z) = aT + \tfrac{1}{2} \log T + \tfrac{1}{2} \log \frac{2a}{\pi} - \frac{1}{8aT} + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big).$$

Also $(N-1)\log(1-qh) = -qT + O(T^2/N)$. Adding the three parts,

$$\log M_F = -L + \tfrac{1}{2} \log T + \log\Big(2\mu_i \sqrt{\frac{2a}{\pi}}\Big) - (q-a)T - \frac{1}{8aT} + O\Big(\frac{1}{T^2} + \frac{T^2}{N}\Big).$$

By Corollary 4.3 with $k = 2$, $\mu(F) = 2\mu_i$ and $\mathrm{tree}(q_F) = a$, we have $\lambda_F = q - a$ and $K_F = 2\mu_i \cdot 2^{3/2} \sqrt{a}\, (4\pi)^{-1/2} = 2\mu_i \sqrt{2a/\pi}$. This is (4.9). Now we compare it with (4.7) at $T = L$. Both expand $\log M_F$, so their difference $(c_F + \frac{1}{8a})/L$ is $O(L^{-2} + L^2/N) = O(L^{-2})$. Letting $N \to \infty$ gives $c_F = -1/(8a)$.

As a check we compute the same value from (B.2). Number the states so that i is state 1. Then $A' = 0$, so in the notation of Appendix A we have $g(z_2) = a^2/z_2$ and $b(z_2) = \mu_i (1 + a/z_2)^2$. At $\alpha = \alpha_F^* = (\frac{1}{2}, \frac{1}{2})$ the radius is $\theta_2 = a$, so $\Psi = a \cos\varphi$ and $\chi = \mu_i a (1 + e^{-\mathrm{i}\varphi})^2 e^{\mathrm{i}\varphi} = 2\mu_i a (1 + \cos\varphi)$. Hence $\mathcal{J} = a$, $\Psi_{111} = 0$, $\Psi_{1111} = a$, $\chi(\alpha_F^*, 0) = 4\mu_i a$, $\chi_1 = 0$ and $\chi_{11} = -2\mu_i a$, and (B.1) gives $c_1 = -\mu_i + \frac{1}{2}\mu_i = -\frac{1}{2}\mu_i$ and $c(\alpha_F^*) = -1/(8a)$. Also $\gamma_F = 0$, because C_F is unchanged when the two states are swapped (Proposition 4.2(b) with $\nu = \mathbf{1}$). So the two values agree.

Part (iv), *the complete graph*. Let $Q = J - dI$, let F have $k \ge 2$ states and let $\mu_i = \mu_0 > 0$ for $i \in F$. Then F satisfies (H), and Corollary 4.3 with $\mu(F) = k\mu_0$ and $\mathrm{tree}(q_F) = k^{k-2}$ gives $\lambda_F = d - k$, α_F^* uniform on F and $K_F = \mu_0 k^{k+1/2} (4\pi)^{-(k-1)/2}$.

We first show that $\gamma_F = 0$, so that $c_F = c(\alpha_F^*)$. By Example 4.11,

$$C_F(\alpha) = \frac{\mu_0 \big(\sum_{i \in F} \sqrt{\alpha_i}\big)^{k/2+1}}{(4\pi)^{(k-1)/2} \prod_{i \in F} \alpha_i^{3/4}},$$

a symmetric function of $(\alpha_i)_{i\in F}$. Take $j\ge 2$ and swap the states 1 and j. This maps the point $\alpha_F^*+v(e_j-e_1)$ to $\alpha_F^*-v(e_j-e_1)$, so $v\mapsto \log C_F(\alpha_F^*+v(e_j-e_1))$ is even and its derivative at $v=0$ vanishes. In the coordinates α' this derivative is the j-th entry of γ_F.

Now we evaluate (B.2) at α_F^* by hand. Put $m=k-1$, the dimension of the torus, and $\varpi=m/k$.

The integrand. At α_F^* we have $\theta(\alpha_F^*)=q-\lambda_F\mathbf{1}$, so every radius is $\theta_i=(d-1)-(d-k)=m$. In the notation of Lemma A.2, $A'=J-I$ of size m, $a_1=b_1=\mathbf{1}$ and $\mu'=\mu_0\mathbf{1}$. With $\mathcal{D}=\operatorname{diag}(z')+I$ the Sherman–Morrison formula gives $X(z')=(\mathcal{D}-\mathbf{1}\mathbf{1}^\top)^{-1}=\mathcal{D}^{-1}+\mathcal{D}^{-1}\mathbf{1}\mathbf{1}^\top\mathcal{D}^{-1}/(1-s)$, where $s=\sum_{i\ge2}1/(1+z_i)$. So $g(z')=s/(1-s)$ and $b(z')=\mu_0(1+g)^2=\mu_0(1-s)^{-2}$. Put $z_i=me^{\mathrm{i}\varphi_i}$ and $\varepsilon_i=e^{\mathrm{i}\varphi_i}-1$, so that $1+z_i=k(1+\varpi\varepsilon_i)$, and put $y_i=\varpi\varepsilon_i/(1+\varpi\varepsilon_i)$ and $Y=\sum_{i\ge2}y_i$. Then $1/(1+z_i)=(1-y_i)/k$, so $1-s=(1+Y)/k$ and $g=k/(1+Y)-1$. Since $(m^2-1)/k=m-1$, the functions of (A.6) are

$$\Psi(\alpha_F^*,\varphi)=\frac{1}{1+Y}+m-1+\varpi\sum_{i\ge2}\varepsilon_i,\qquad \chi(\alpha_F^*,\varphi)=\mu_0k^2m^m\,(1+Y)^{-2}e^{\mathrm{i}\sum_{i\ge2}\varphi_i}.$$

Taylor coefficients. Write $\mathcal{S}_j=\sum_{i\ge2}\varphi_i^j$. Expanding $\varepsilon_i=\mathrm{i}\varphi_i-\frac12\varphi_i^2-\frac{\mathrm{i}}{6}\varphi_i^3+\frac{1}{24}\varphi_i^4+O(\varphi_i^5)$ in $y_i=\varpi\varepsilon_i-\varpi^2\varepsilon_i^2+\varpi^3\varepsilon_i^3-\varpi^4\varepsilon_i^4+O(\varepsilon_i^5)$ gives

$$y_i=\mathrm{i}\varpi\varphi_i+\varpi(\varpi-\tfrac12)\varphi_i^2+\mathrm{i}\varpi(\varpi-\varpi^2-\tfrac16)\varphi_i^3+\varpi(\tfrac{1}{24}-\tfrac{7}{12}\varpi+\tfrac32\varpi^2-\varpi^3)\varphi_i^4+O(\varphi_i^5).$$

We put this into $1/(1+Y)=1-Y+Y^2-Y^3+Y^4-\cdots$ and into $\log(\chi/\chi(\alpha_F^*,0))=-2\log(1+Y)+\mathrm{i}\mathcal{S}_1$. The linear term of Ψ cancels, and with $\mathcal{P}_j$ the part of degree j of the Taylor polynomial of $\Psi(\alpha_F^*,\cdot)$ and $\mathcal{L}_j$ that of $\log(\chi/\chi(\alpha_F^*,0))$,

$$\begin{aligned}
\mathcal{P}_2&=-\varpi^2(\mathcal{S}_2+\mathcal{S}_1^2),\\
\mathcal{P}_3&=\mathrm{i}\big[\varpi^2(\varpi-1)\mathcal{S}_3+\varpi^2(2\varpi-1)\mathcal{S}_1\mathcal{S}_2+\varpi^3\mathcal{S}_1^3\big],\\
\mathcal{P}_4&=\varpi^2\big[(\tfrac{7}{12}-\tfrac32\varpi+\varpi^2)\mathcal{S}_4+(\varpi-\tfrac12)^2\mathcal{S}_2^2+(\tfrac13-2\varpi+2\varpi^2)\mathcal{S}_1\mathcal{S}_3\\
&\qquad+3\varpi(\varpi-\tfrac12)\mathcal{S}_1^2\mathcal{S}_2+\varpi^2\mathcal{S}_1^4\big],\\
\mathcal{L}_1&=\mathrm{i}(1-2\varpi)\mathcal{S}_1,\qquad \mathcal{L}_2=\varpi(1-2\varpi)\mathcal{S}_2-\varpi^2\mathcal{S}_1^2.
\end{aligned}$$

The formula as a Gaussian mean. By the proof of Lemma B.2, $c_1=\mathbb{E}\mathcal{Q}_2$ with $\mathcal{Q}_2=a_0(\mathcal{P}_4+\frac12\mathcal{P}_3^2)+A_1\mathcal{P}_3+A_2$, where $a=\chi$, A_j is the part of degree j of its Taylor polynomial, and the variable is a centered Gaussian vector x with covariance $\mathcal{J}^{-1}$. Since $A_1=a_0\mathcal{L}_1$ and $A_2=a_0(\mathcal{L}_2+\frac12\mathcal{L}_1^2)$,

$$c(\alpha_F^*)=\frac{c_1}{a_0}=\mathbb{E}\Big[\mathcal{P}_4+\mathcal{L}_2+\tfrac12(\mathcal{P}_3+\mathcal{L}_1)^2\Big](x).$$

Here $\mathcal{P}_2=-\frac12x^\top\mathcal{J}x$ gives $\mathcal{J}=2\varpi^2(I+\mathbf{1}\mathbf{1}^\top)$, of size m, and $\mathcal{J}^{-1}=\sigma_0(I-\mathbf{1}\mathbf{1}^\top/k)$ with $\sigma_0=1/(2\varpi^2)$.

An exchangeable Gaussian vector. Let $G_1,\dots,G_k$ be independent standard Gaussian variables, $\bar G$ their mean and $w_i=G_i-\bar G$. Then $\sum_{i=1}^k w_i=0$ and $\mathbb{E}[w_iw_j]=\delta_{ij}-1/k$, so $(x_2,\dots,x_k)$ has the law of $\sqrt{\sigma_0}(w_2,\dots,w_k)$. Put $W_j=\sum_{i=1}^k w_i^j$. Then $\mathcal{S}_1=-\sqrt{\sigma_0}\,w_1$ and $\mathcal{S}_j=\sigma_0^{j/2}(W_j-w_1^j)$ for $j\ge2$. We put these into the Taylor coefficients and use $\sigma_0\varpi^2=\frac12$, $1-\varpi=1/k$ and $2\varpi-1=(k-2)/k$. The terms in w_1^3 cancel in $\mathcal{P}_3+\mathcal{L}_1$ and the terms in w_1^4 cancel in $\mathcal{P}_4$, and

$$\begin{aligned}
\mathcal{P}_3+\mathcal{L}_1&=-\frac{\mathrm{i}\sqrt{\sigma_0}}{2k}\big[W_3+(k-2)w_1(W_2-2)\big],\\
\mathcal{P}_4&=\frac{1}{4\varpi^2}\Big[(\tfrac{7}{12}-\tfrac32\varpi+\varpi^2)W_4+(\varpi-\tfrac12)^2W_2^2\\
&\qquad+(\varpi+1)(\varpi-\tfrac12)w_1^2W_2-(\tfrac13-2\varpi+2\varpi^2)w_1W_3\Big],\\
\mathcal{L}_2&=\frac{1-2\varpi}{2\varpi}(W_2-w_1^2)-\frac12w_1^2.
\end{aligned}$$

The moments. By exchangeability and $\sum_i w_i = 0$, every symmetric function Υ of w has $\mathbb{E}[w_1 \Upsilon(w)] = \frac{1}{k}\mathbb{E}[\Upsilon(w)\sum_i w_i] = 0$. So $\mathbb{E}[w_1 W_3] = 0$ and $\mathbb{E}[w_1 W_3 W_2] = 0$. The vector w is a standard Gaussian vector in the hyperplane $\sum_i w_i = 0$, so W_2 has the χ^2 law with m degrees of freedom and is independent of $w/\sqrt{W_2}$. Hence $\mathbb{E}[W_2^j] = m(m+2)\cdots(m+2j-2)$ and $\mathbb{E}[w_1^2 W_2^j] = \mathbb{E}[W_2^{j+1}]/k$, since $\mathbb{E}[w_1^2 \mid W_2] = W_2/k$ by symmetry. By Wick’s formula, $\mathbb{E}[w_1^4] = 3\varpi^2$, $\mathbb{E}[w_1^6] = 15\varpi^3$ and $\mathbb{E}[w_1^3 w_2^3] = 9\varpi^2(-\frac{1}{k}) + 6(-\frac{1}{k})^3$, so $\mathbb{E}[W_4] = 3k\varpi^2$ and $\mathbb{E}[W_3^2] = 15k\varpi^3 - k(k-1)(9\varpi^2/k + 6/k^3) = 6m(m-1)/k$. With these values,

$$\begin{aligned}
\mathbb{E}[\mathcal{P}_4] &= \frac{3k}{4}\Big(\frac{7}{12} - \frac{3}{2}\varpi + \varpi^2\Big) + \frac{m(m+2)}{4\varpi^2}\Big(\varpi - \frac{1}{2}\Big)\Big(\varpi - \frac{1}{2} + \frac{\varpi+1}{k}\Big) \\
&= \frac{m^4 + 6m^3 - m^2 - 2}{16mk}, \\
\mathbb{E}[\mathcal{L}_2] &= \frac{1-2\varpi}{2\varpi}\, m\varpi - \frac{\varpi}{2} = -\frac{m^2}{2k}, \\
\tfrac{1}{2}\mathbb{E}[(\mathcal{P}_3 + \mathcal{L}_1)^2] &= -\frac{\sigma_0}{8k^2}\Big(\mathbb{E}[W_3^2] + (k-2)^2\mathbb{E}\big[w_1^2(W_2-2)^2\big]\Big) \\
&= -\frac{\sigma_0}{8k^2}\cdot\frac{m(m-1)}{k}\big(m^3 + m^2 + 2m + 2\big) = -\frac{m^4 + m^2 - 2}{16mk},
\end{aligned}$$

where we used $\mathbb{E}[w_1^2(W_2-2)^2] = m(m^2+2m+4)/k$ and $\sigma_0 = k^2/(2m^2)$. Adding the three,

$$c_F = c(\alpha_F^*) = \frac{(m^4 + 6m^3 - m^2 - 2) - 8m^3 - (m^4 + m^2 - 2)}{16mk} = -\frac{2m^2(m+1)}{16mk} = -\frac{k-1}{8}.$$

For $k = 2$ this is the value $-\frac{1}{8}$ of the pair. □

References

[1] F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl and M. A. McClain (eds.), *NIST Digital Library of Mathematical Functions*, https://dlmf.nist.gov/.